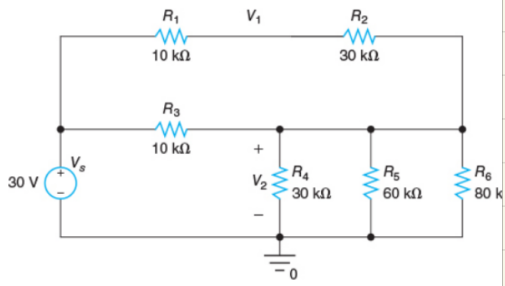
1. Consider the following circuit:



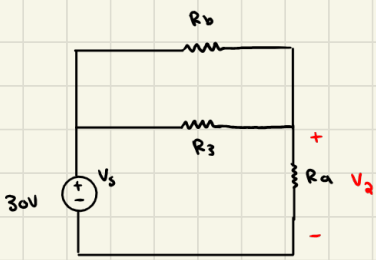
a. Calculate equivalent resistance.
b. Calculate the voltage across all resistances and current flowing through all resistances.
c. Calculate V_1 and V_2.

begin by combining the circuit in series and parallel to find R_{eq}

$$\boxed{R_{eq} = 24\ k\Omega}$$

$$R_a = \left(\frac{1}{R_4} + \frac{1}{R_5} + \frac{1}{R_6}\right)^{-1} \Rightarrow \left(\frac{1}{30} + \frac{1}{60} + \frac{1}{80}\right)^{-1} \Rightarrow R_a = 16\ k\Omega$$

$$R_b = R_1 + R_2 \Rightarrow 10 + 30 \Rightarrow R_b = 40\ k\Omega$$

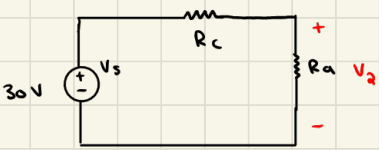


$$R_c = \left(\frac{1}{R_b} + \frac{1}{R_3}\right)^{-1} \Rightarrow \left(\frac{1}{40} + \frac{1}{10}\right)^{-1}$$

$$R_c = 8\ k\Omega$$

$$V_2 = \frac{R_a}{R_a + R_c} \cdot V_s \Rightarrow \frac{16}{16 + 8} \cdot 30$$

$$\boxed{V_2 = 20V}$$



$$V_{R_c} = \frac{R_c}{R_c + R_a} \cdot V_s \Rightarrow \frac{8}{8 + 16} \cdot 30$$

$$V_{R_c} = 10V$$

- if we already have got the V_{R_c} is 10V and R_c consists of R_b and R_3 that would mean R_b and R_3 have a voltage of 10V

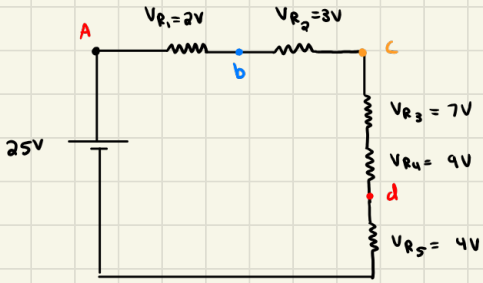
Made with Goodnotes

$$I_{R_b} = \frac{V_{R_c}}{R_b} = \frac{10}{40000} = \boxed{0.25\ mA}$$

R_b consists of R_1 and R_2 they are both in series which means they have the same current going through them so now we can get the voltage of R_2

$$V_{R_2} = I_{R_b} R_2 \Rightarrow (0.25 \times 10^{-3})(30 \times 10^{3}) = 7.5V$$

New technique:



- at Point a the voltage will be 25V
- at Point b we can count the right side voltages to get b this is completly fine because what we are doing is adding voltage drops

So: $V_b =$ (sum of all voltage rise and drops from ground to b)

$V_b = 23V$

- at Point c: $V_c =$ (sum of all voltage rise and drops from gnd to c)

$V_c = 20V$

Made with Goodnotes

- at Point d : V_d = (Sum of all voltage rise and drop from gnd to to d)

$V_d = 4V$

we introduce this technique because we already got V_{R_2} and we have V_2 we need to get V_1 we can now use the new technique we learned to get V_1

$$V_1 = V_{R_2} + V_2 \Rightarrow 7.5 + 20$$

$$\boxed{V_1 = 27.5\ V}$$

$$I_{R_b} = I_{R_1} = I_{R_2} = 0.25mA$$

- because they are in series so they have the same current

$$I_{R_4} = \frac{V_2}{R_4} \Rightarrow \frac{20}{30000} = 0.667\ mA$$

$$I_{R_5} = \frac{V_2}{R_5} \Rightarrow \frac{20}{60000} = 0.333\ mA$$

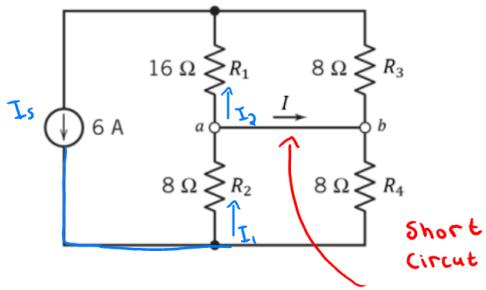
$$I_{R_6} = \frac{V_2}{R_6} \Rightarrow \frac{20}{80000} = 0.25\ mA$$

$$I_{R_3} = \frac{V_{R_c}}{R_3} = \frac{10}{10000} = 1mA$$

Made with Goodnotes

2. Find I and V_{ab} in the following circuits.

a.



the reason why I_1 and I_2 go up is because I_s is supplying current going down

when you have short circut the resistance is zero so for V_{ab} its going to be zero

- we can apply current divider rule to get I_1 and I_2

$$I_1 = \frac{\frac{1}{R_2}}{\frac{1}{R_2} + \frac{1}{R_4}} \cdot I_s \Rightarrow \frac{\frac{1}{8}}{\frac{1}{8} + \frac{1}{8}} \cdot 6 \Rightarrow 3\,mA$$

$$I_2 = \frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_3}} \cdot I_s \Rightarrow \frac{\frac{1}{16}}{\frac{1}{16} + \frac{1}{8}} \cdot 6 \Rightarrow 2\,mA$$

we can apply a KCL to get I:

entering = exiting

$$I_1 = I_2 + I$$

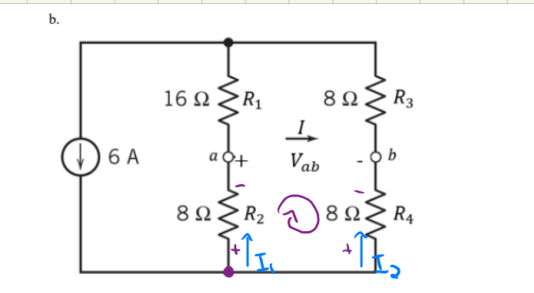
$$I_1 - I_2 = I$$

$$3 - 2 = I$$

$$\boxed{I = 1\,A}$$

Made with Goodnotes

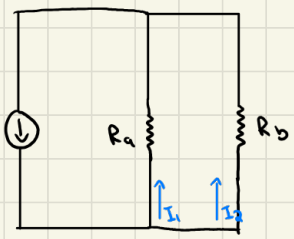
b.



here I is zero because the electrons cannot jump to the other side so $I = 0$

we will use current divider rule to find I_1 and I_2 because we want to apply a KVL to find the V_{ab} notice not all of the resistors are in parallel so you need to combine them so they are all in parallel but the current will stay the same be they are in series

$$R_a = R_1 + R_2 \Rightarrow 16 + 8 = 24\ \Omega$$
$$R_b = R_3 + R_4 \Rightarrow 8 + 8 = 16\ \Omega$$



$$I_1 = \frac{\frac{1}{R_a}}{\frac{1}{R_a} + \frac{1}{R_b}} \cdot I_s \Rightarrow \frac{\frac{1}{24}}{\frac{1}{24} + \frac{1}{16}} \cdot 6$$

$$\boxed{I_1 = 2.4\ A}$$

$$I_2 = \frac{\frac{1}{R_b}}{\frac{1}{R_b} + \frac{1}{R_a}} \cdot I_s \Rightarrow \frac{\frac{1}{16}}{\frac{1}{16} + \frac{1}{24}} \cdot 6$$

$$\boxed{I_2 = 3.6\ A}$$

$$V_{R_2} + V_{ab} - V_{R_4} = 0$$

$$V_{ab} = V_{R_4} - V_{R_2}$$

$$V_{ab} = I_1 R_4 - I_2 R_2$$

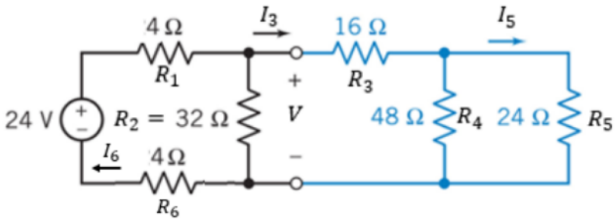
$$V_{ab} = (2.4)(8) - (3.6)(8)$$

$$\boxed{V_{ab} = -9.6\ V}$$

- the negitive means we are going the other way

Made with Goodnotes

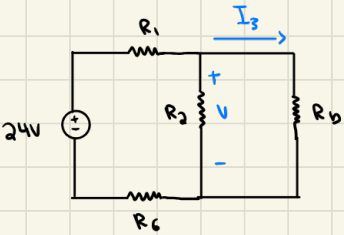
3. Find V, I_3, I_5, and I_6.



begin by combining the resistors in series and in Parallel:

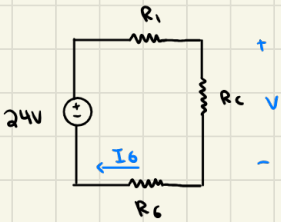
$$R_a = \left(\frac{1}{R_4} + \frac{1}{R_5}\right)^{-1} \Rightarrow \left(\frac{1}{48} + \frac{1}{24}\right)^{-1} = 16\ \Omega$$

$$R_b = R_3 + R_a \Rightarrow 16 + 16 = 32\ \Omega$$



$$R_c = \left(\frac{1}{R_2} + \frac{1}{R_b}\right)^{-1} \Rightarrow \left(\frac{1}{32} + \frac{1}{32}\right)^{-1}$$

$$R_c = 16\ \Omega$$



$$I_6 = \frac{V}{R_1 + R_2 + R_c} = \frac{24}{4+4+16} = 1\ A$$

$$\boxed{I_6 = 1A}$$

$$V = I_6 R_c \Rightarrow (1)(16)$$

$$\boxed{V = 16V}$$

$$I_3 = \frac{V}{R_b} = \frac{16}{32} = 0.5\ A$$

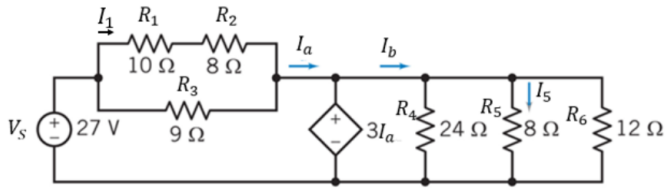
$$\boxed{I_3 = 0.5\ A}$$

Made with Goodnotes

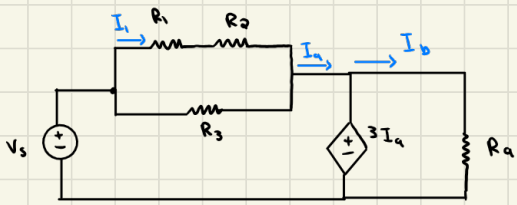
$$I_5 = \frac{\frac{1}{R_5}}{\frac{1}{R_5}+\frac{1}{R_4}} \cdot I_3 \Rightarrow \frac{\frac{1}{24}}{\frac{1}{24}+\frac{1}{48}} \cdot 0.5$$

$$\boxed{I_5 = 0.333\ A}$$

4. Find I_1, I_a, I_b, and I_5.



$$R_a = \left(\frac{1}{R_4}+\frac{1}{R_5}+\frac{1}{R_6}\right)^{-1} \Rightarrow \left(\frac{1}{24}+\frac{1}{8}+\frac{1}{12}\right)^{-1} \Rightarrow R_a = 4\,\Omega$$

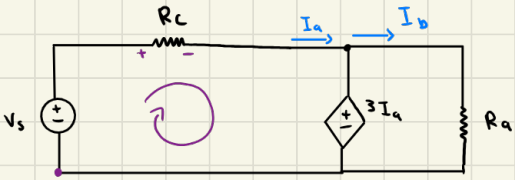


$$R_b = R_1 + R_2 \Rightarrow 10 + 8 \Rightarrow$$

$$R_b = 18\,\Omega$$

$$R_c = \left(\frac{1}{R_b}+\frac{1}{R_3}\right)^{-1} \Rightarrow \left(\frac{1}{18}+\frac{1}{9}\right)^{-1}$$

$$R_c = 6\,\Omega$$



apply KVL:

$$-V_s + V_{R_c} + 3I_a = 0$$

$$I_a R_c + 3I_a = V_s$$

$$I_a (R_c + 3) = V_s$$

$$I_a = \frac{27}{6+3} \Rightarrow \boxed{I_a = 3\ A}$$

Made with Goodnotes

we have a current controlled voltage source that is controlled by I_a we already found I_a so we can find that CCVS and we notice that its parallel to R_a which has current I_b going through it so we can find I_b after finding CCVS

$$CCVS = 3\,I_a \Rightarrow (3)(3) \Rightarrow CCVS = 9V$$

$$I_b = \frac{CCVS}{R_a} \Rightarrow \frac{9}{4} \Rightarrow \boxed{I_b = 2.25\ A}$$

looking at the original we notice that we can apply a current divider rule to solve for I_5:

$$I_5 = \frac{\frac{1}{R_5}}{\frac{1}{R_5}+\frac{1}{R_4}+\frac{1}{R_6}} \cdot I_b \Rightarrow \frac{\frac{1}{8}}{\frac{1}{8}+\frac{1}{24}+\frac{1}{12}} \cdot 2.25 \Rightarrow \boxed{I_5 = 1.125\,A}$$

to get I_1 we can find the voltage going through R_c because we already know the current (I_a)

$$V_{R_c} = I_a R_c \Rightarrow (3)(6) \Rightarrow V_{R_c} = 18V$$

$$V_{R_c} = V_{R_b} = V_{R_3}$$

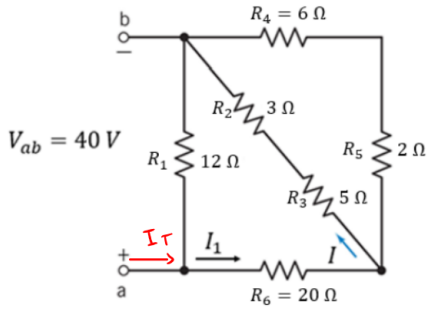
- R_b consists of R_1 and R_2 in series so if we find the current going R_b it will be the same for R_1 and R_2 because they are in series

$$I_1 = \frac{V_{R_c}}{R_b} = \frac{18}{18}$$

$$\boxed{I_1 = 1A}$$

Made with Goodnotes

5. Find I and I_1.

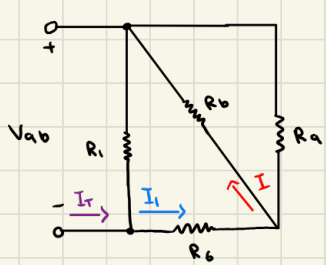


Combine the resistor in series or in Parallel then begin to solve

$R_a = R_4 + R_5 \Rightarrow 6 + 2 \Rightarrow R_a = 8\,\Omega$

$R_b = R_2 + R_3 \Rightarrow 3 + 5 \Rightarrow R_b = 8\,\Omega$

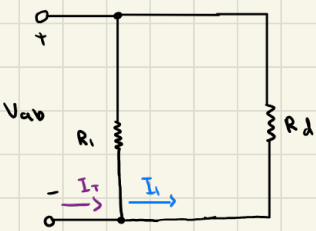
1)



$$R_c = \left(\frac{1}{R_b} + \frac{1}{R_a}\right)^{-1} \Rightarrow \left(\frac{1}{8} + \frac{1}{8}\right)^{-1}$$

$$R_c = 4\,\Omega$$

2)



$R_d = R_c + R_6 \Rightarrow 4 + 20 \Rightarrow R_d = 24\,\Omega$

$$R_e = \left(\frac{1}{R_1} + \frac{1}{R_d}\right)^{-1} \Rightarrow \left(\frac{1}{12} + \frac{1}{24}\right)^{-1}$$

$$R_e = 8\,\Omega$$

$$I_T = \frac{V_{ab}}{R_e} = \frac{40}{8} \Rightarrow I_T = 5\,A$$

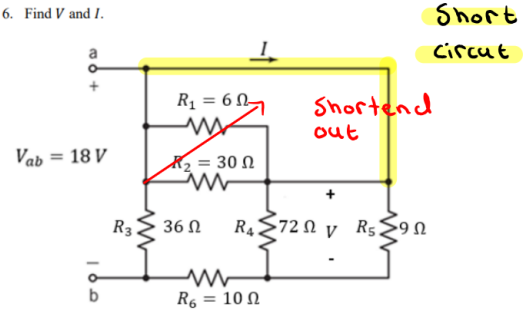
now that we have got I_T we can apply a current divider rule to find I_1 we are looking at 2)

$$I_1 = \frac{\frac{1}{R_d}}{\frac{1}{R_d} \cdot \frac{1}{R_1}} \cdot I_T \Rightarrow \frac{\frac{1}{24}}{\frac{1}{24} + \frac{1}{12}} \cdot 5 \Rightarrow \boxed{I_1 = 1.6\,A}$$

Made with Goodnotes

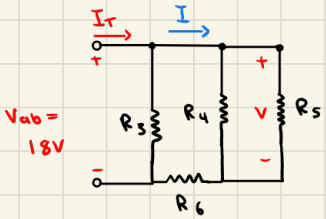
$$I = \frac{\frac{1}{R_b}}{\frac{1}{R_b} + \frac{1}{R_a}} \cdot I_1 \Rightarrow \frac{\frac{1}{8}}{\frac{1}{8} + \frac{1}{8}} \cdot 1.6 \qquad \boxed{I = 0.833\ A}$$

6. Find V and I.



R_1 and R_2 are parallel to a short circut so they will get shortend out

1)

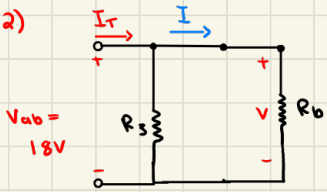


- we want to apply current divider rule here to get I but the problem is that we don't have a main current so what will do is set I_T in the start and sum the circut up

$$R_a = \left(\frac{1}{R_4} + \frac{1}{R_5}\right)^{-1} \Rightarrow \left(\frac{1}{72} + \frac{1}{9}\right)^{-1} \Rightarrow R_a = 8\Omega$$

$$R_b = R_a + R_6 \Rightarrow 8 + 10 \Rightarrow R_b = 18\Omega$$

2)

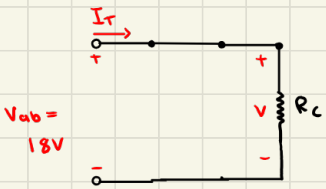


$$R_c = \left(\frac{1}{R_b} + \frac{1}{R_3}\right)^{-1} \Rightarrow \left(\frac{1}{18} + \frac{1}{36}\right)^{-1}$$

$$R_c = 12\Omega$$

Made with Goodnotes

3)



$$I_T = \frac{V_{ab}}{R_c} = \frac{18}{12} \Rightarrow I_T = 1.5\ A$$

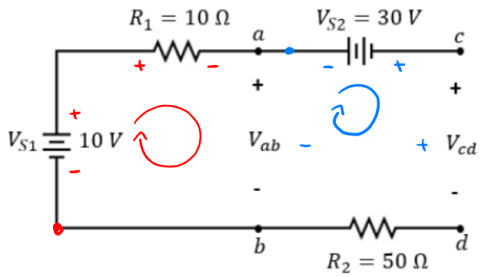
now we can apply current divider rule because now we have a main current supply we are now looking at 2) we won't use 1) for the current divider rule because we don't know where the current is going but using 2) we are sure it goes into R_b

$$I = \frac{\frac{1}{R_b}}{\frac{1}{R_b} + \frac{1}{R_3}} \cdot I_T \Rightarrow \frac{\frac{1}{18}}{\frac{1}{18} + \frac{1}{36}} \cdot 1.5 \qquad \boxed{I = 1A}$$

to get V we will use R_b because they are both parallel and they would have the same voltage and we can also be sure that I actually goes through it

$$V = I R_b = (1)(8) = \boxed{V = 8V}$$

7. Find V_{ab} and V_{cd} for the following network.



Made with Goodnotes

$-V_{S1} + V_{R_1} + V_{ab} = 0$

$-V_{S1} + I_1 R_1 + V_{ab} = 0$

$-10 + 0 R_1 = -V_{ab}$

$\boxed{V_{ab} = 10\ V}$

- the reasoning why I_1 is zero is because it is a open circut so we cannot have current when we have a open circut the electrons cannot jump from one terminal to the other

$-V_{ab} + V_{S2} + V_{cd} + V_{R_2} = 0$

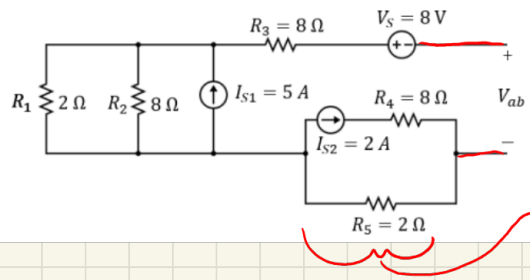
$-V_{ab} + V_{S2} + V_{cd} + I_2 R_2 = 0$

$-10 + 30 + 0 R_2 = -V_{cd}$

$\boxed{V_{cd} = 20V}$

- the reasoning why I_2 is zero is because we have a open circut when you have a open circut electrons cannot jump from one terminal to the other

8. Find V_{R2}, I_{R2}, V_{R3}, I_{R3}, V_{R5}, I_{R5}, and V_{ab}.



voltage is the same for both

We can find I_{R2} by applying current divider rule:

$$I_{R2} = \frac{\frac{1}{R_2}}{\frac{1}{R_2} + \frac{1}{R_1}} \cdot I_{S1} \Rightarrow \frac{\frac{1}{8}}{\frac{1}{8} + \frac{1}{2}} \cdot 5 \Rightarrow \boxed{I_{R2} = 1A}$$

$$V_{R2} = I_{R_2} \cdot R_2 \Rightarrow (1)(8) \Rightarrow \boxed{V_{R2} = 8V}$$

Made with Goodnotes

R_3 is part of a open circut there will not be any current going through it so:

$$\boxed{I_{R_3} = 0\ A}$$

$$\boxed{V_{R_3} = 0V}$$

Notice that we have a dependent current source that will give us a constant current no matter what happens so:

$$I_{R_5} = I_{S_2} = 2A \quad \text{So,} \quad \boxed{I_{R_5} = 2A}$$

$$V_{R_5} = I_{R_5} R_5 \Rightarrow (2)(2)$$

$$\boxed{V_{R_5} = 4V}$$

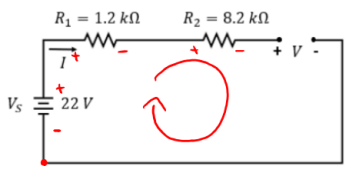
Finding V_{ab}: $V_{ab} = V_s - V_{R_5}$

$$V_{ab} = 8 - 4$$

$$\boxed{V_{ab} = 4V}$$

9. Find V and I for the following networks.

a.



When you have a circut that looks like this that has a gap current cannot pass so there will be no current

$$\boxed{I = 0\ A}$$

$$-V_s + V_{R_1} + V_{R_2} + V = 0$$

$$-V_s + I_{R_1} R_1 + I_2 R_2 + V = 0$$

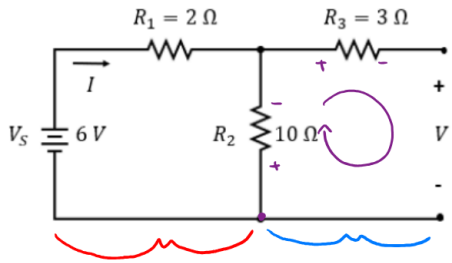
$$-22 + (0)(2) + (0)(3) = -V$$

$$\boxed{22V = V}$$

☆ electrons cannot jump from one terminal to the other

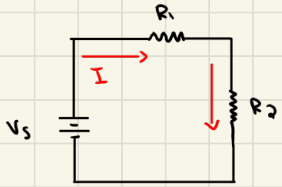
Made with Goodnotes

b.



- current is flowing here
- there is no current flowing here its a open circut

you can view this circut like this:



- because the current gets to R_3 and cannot continue so it actually moves in a loop

$$I = \frac{V_s}{R_1 + R_2} \Rightarrow \frac{6}{2+10} \Rightarrow \boxed{I = 0.5A}$$

$$+V_{R2} + V_{R3} + V = 0$$

$$IR_2 + IR_3 = -V$$

$$(0.5)(10) + (0)(3) = -V$$

$$\boxed{V = -5V}$$

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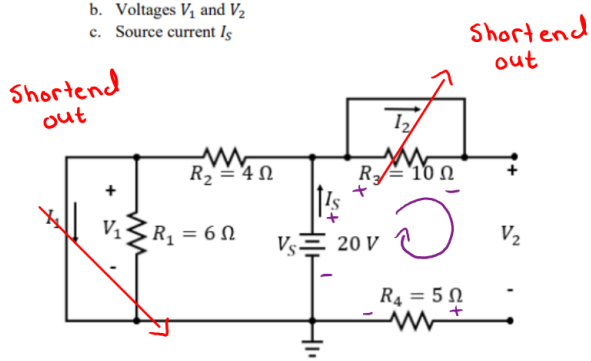
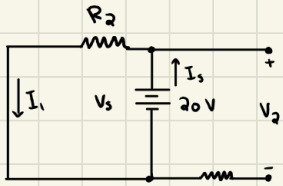
10. For the network shown below, calculate:

a. Currents I_1 and I_2
b. Voltages V_1 and V_2
c. Source current I_S

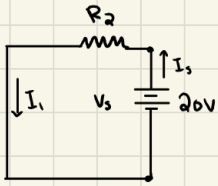




- I_2 is going to be zero because on the right side you have a open circut but you can find I_1 because the current is flowing in a closed circut

$$\boxed{I_2 = 0\ A}$$

you can look at this circut like this:



you can draw the circut like this because the right side is a open circut and the electrons would get to R_4 and not do a full circular path

- everything in this circut is in series so I_1 is the same as I_s

$$I_1 = \frac{V_s}{R_2} = \frac{20}{4} \Rightarrow \boxed{I_1 = 5A}$$

$$\boxed{I_s = 5A}$$

apply KVL:

$$-V_s + V_{R_3} + V_2 + V_{R_4} = 0$$

$$-V_s + I_3 R_3 + V_2 + I_4 R_4 = 0$$

$$-20 + (0)(5) + V_2 + (0)(5) = 0$$

$$\boxed{V_2 = 20V}$$

- when a resistor is shortend out it has a resistance of zero so:

$$V_1 = I_1 R_1 \Rightarrow V_1 = I_1 (0)$$

$$\boxed{V_1 = 0\ V}$$

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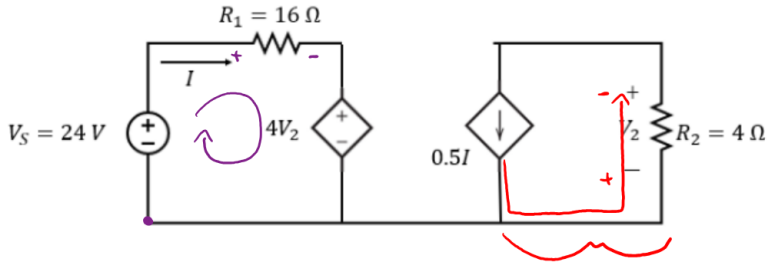
11. Find V_2 and I.



- have the same voltage because they are parallel

- also notice that the current dependent current source tells us that we will go from the bottom of the resistor to top so by that we should have a + on the bottom and - on top as current goes from high potential to low potential and to show this movement in our equation we need to add a negitive sign to it

$$V_2 = -0.5I \cdot R_2 \Rightarrow V_2 = -0.5I(4)$$

$$V_2 = -2I$$

apply KVL:

$$-V_s + V_{R_1} + 4V_2 = 0$$

$$-V_s + IR_1 + 4(-2I) = 0$$

$$I(16) - 8I = 24$$

$$I(8) = 24$$

$$I = \frac{24}{8}$$

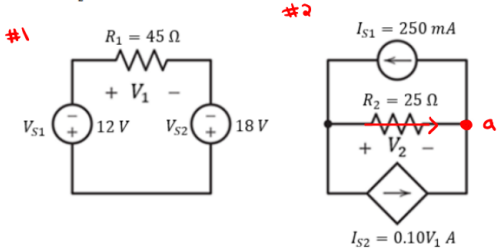
$$\boxed{I = 3A}$$

$$V_2 = -2(3)$$

$$\boxed{V_2 = -6\ V}$$

Made with Goodnotes

12. Find V_2.



#1

V_1 = high Potential − low Potential

$V_1 = 18 - 12$

$V_1 = 6V$

#2

$I_{S2} = 0.10\ (6)$

$I_{S2} = 0.6\ A$

$V_2 = I_2 R_2 \Rightarrow (-0.35)(25)$

$\boxed{V_2 = -8.75\ V}$

• apply a KCL at node a)

entering = exiting

$I_{S2} + I_2 = I_{S1}$

$I_2 = I_{S1} - I_{S2}$

$I_2 = 0.25 - 0.6$

$I_2 = -0.35$

Made with Goodnotes